

- ★ GL-TURBO Global Headquarters
- GL-TURBO Global Branches



GL-TURBO

USA · China · Denmark · Germany · India

www.glturbo.com
info@glturbo.com

GL-TURBO (GPAC-2401)_US_Headquarter 04 | 2024

GPAC®

PRECISION AERATION CONTROL SYSTEM



GL-TURBO

CATALOG

ABOUT GPAC	P02
BIOCHEM HISTORY	P03
CONTROL PHILOSOPHY	P04
SYSTEM COMPOSITION	P05
BLOWER CONTROL	P07
DYNAMIC DO SETPOINTS	P09
MANUAL DO SETPOINTS	P10
VALVE CONTROL MODULE (MOV)	P11
PRECISION DOSING	P12
INTELLECTUAL PROPERTY	P13
NOTABLE INSTALLTIONS	P14



ABOUT GPAC

Since its establishment, GL-TURBO® has been committed to the research, development, and promotion of the GPAC® precise aeration control system for blowers, achieving fundamental breakthroughs in system stability and control accuracy.

In 2022, GL-TURBO® acquired BioChem Technology, Inc. (BCTi), a US-based company founded in 1978. BCTi was one of the earliest companies in the world to develop and promote precise aeration control technology. With this strong alliance, GPAC® has elevated precise aeration control technology to a new level.

GPAC®'s precise aeration control systems have been widely applied in hundreds of projects across more than 20 major countries and regions, including the United States, Singapore, and China. These systems operate stably, with evident control and energy-saving effects, earning the trust and appreciation of a wide range of customers

BIOCHEM HISTORY

- 1977
- Founded in Philadelphia as a consulting first for chemical / pharmaceutical process.
- 1995
- Pivoted to waste water treatment processes based on the expertise in biological fermentation processes developed by famous scientist George Lee.
- 1996
- Famous entrepreneur Lewis Somers acquired majority control of BioChem with the sole mandate to develop the best wastewater control technologies and products.
- 2004
- Developed the world's first precision aeration control system at Enfield WPCF in Connecticut, USA.
- 2005
- Implemented China's earliest precision aeration control systems at Zhuyuan and Bailonggang WWTPs in Shanghai.
- 2010
- Patented the improved "Bioprocess Aeration Control System" in USA, EU and China.
- 2014
- Patented "Low drift ion selective electrode sensors" for an innovative and more stable NH4 and NO3 sensors in USA, EU and China.
- 2022
- Joined GL-TURBO group and moved HQ to Boston.

CONTROL PHILOSOPHY

GPAC® pioneered a brand-new control concept in the wastewater treatment industry, utilizing feedforward control based on the International Water Association (IWA) Activated Sludge Model (ASM).

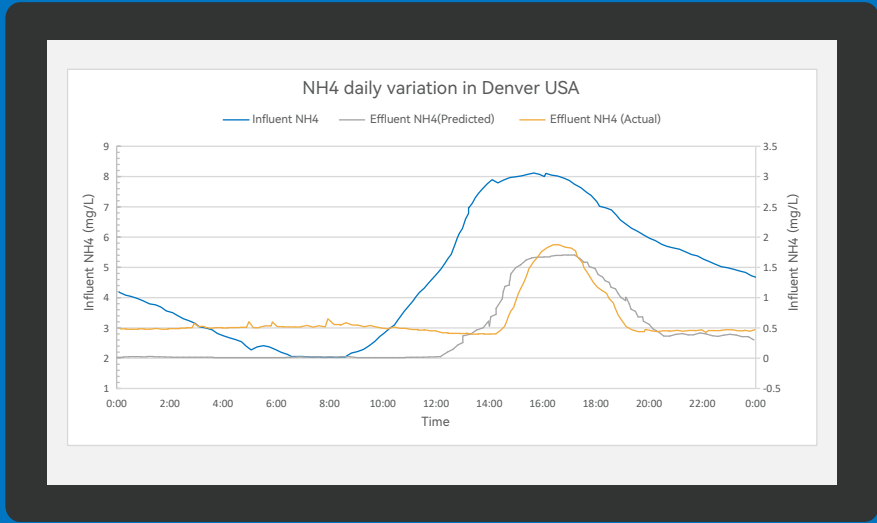
GPAC® accumulated extensive professional knowledge about activated sludge during the early stages of its bio-fermentation process consulting, gaining a deep understanding of the complex ASM model and finding ways to simplify it.

$$OUR = SOTR \cdot \left(\frac{\beta \cdot C_s - DO}{C_{s,20}} \right) \cdot 1.024^{T-20} \cdot \alpha \cdot F$$

Based on this foundation, considering various factors such as the actual operating conditions of WWTPs, changes in influent water volume and quality, and simulating the use of multiple combinations of return ratios, blower air volumes, and valve openings, the system predicts the corresponding effluent water quality and total flow required. According to the simulation results, an optimal combination is selected to minimize the total flow and energy consumption while ensuring effluent compliance. The blowers and valves are then adjusted accordingly, and the adjustment effect is checked. The entire process generally takes 12-15 minutes.

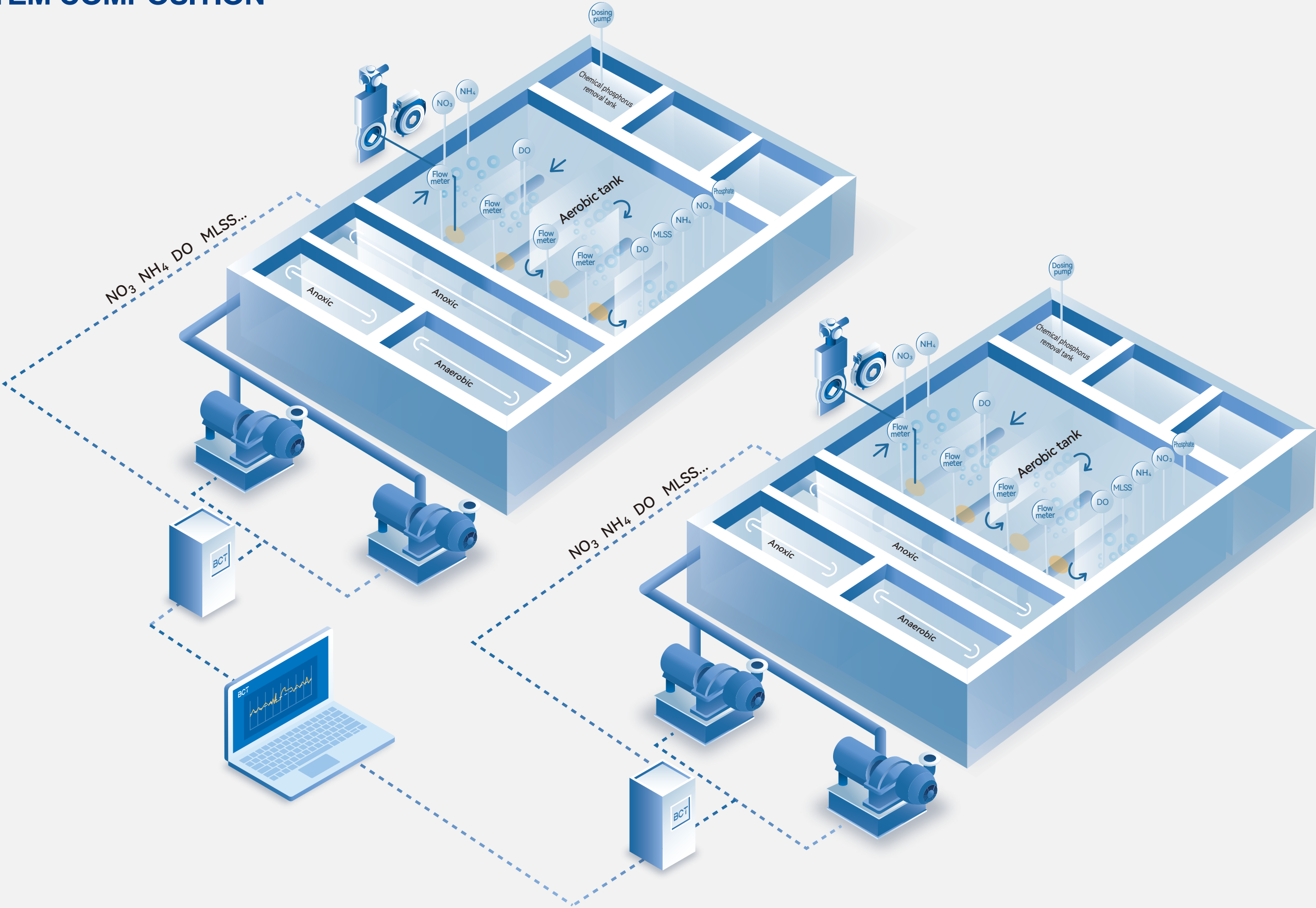
The feedforward control of GPAC®'s precise aeration control system effectively avoids various problems of traditional PI feedback control, such as poor control effect, system instability, too frequent adjustments, and short service life of hardware such as blowers and valves.

GPAC®'s precise aeration control system only needs to be adjusted once every 12-15 minutes and can maintain a control accuracy of over 90%.



The actual effluent ammonia nitrogen concentration is very close to GPAC®'s predicted value.

SYSTEM COMPOSITION



BLOWER CONTROL

The blower is the most important hardware component of GPAC®'s precise aeration control system. Precise aeration cannot be achieved without the support of a good blower.

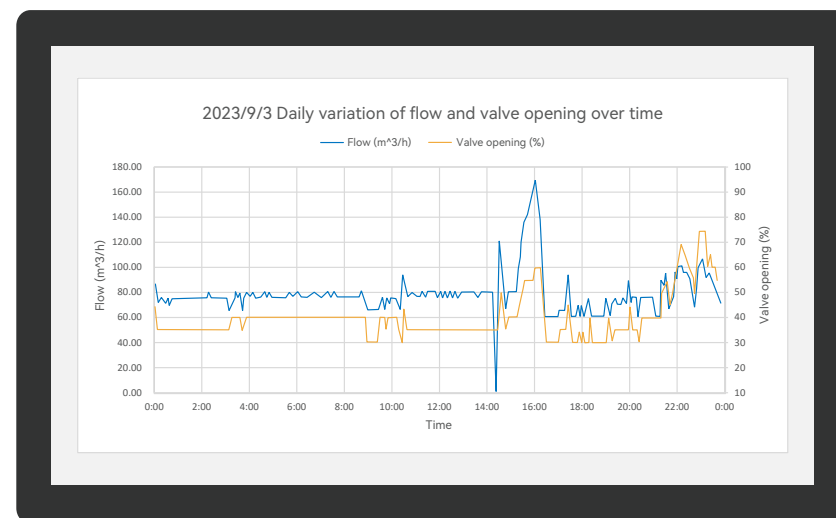
Flow control

Based on an in-depth understanding of blowers and the requirements of the precise aeration control system for accurate flow distribution, GPAC® cooperated with blower manufacturers to first advocate and use flow control of blowers. The flow setpoint (instead of the traditional pressure setpoint) is sent to the blower's main control cabinet (MCP), and the MCP adjusts the opening of each blower accordingly to precisely provide the total flow required by the system.

This flow control mode solves a long-standing problem of pressure control, where small pressure changes in the pipeline can cause large variations in the flow of the blower.

Accurate flow distribution

GPAC® continuously explores the impact of blower and valve performance on the precise aeration control system and has now raised the accurate distribution of flow to a new level.



From Wuxi Gaoxin WWTP II
The entire system has 20 tanks, and during the displayed period, GPAC® precisely controlled the average flow of each tank to less than 75m³/h (1.25m³/min).

Blower Types

GPAC®'s precise aeration control system supports all types of blowers, including single-stage centrifugal, air bearing, maglev, multi-stage centrifugal, roots, and screw blowers.

Different types of blowers offer varying levels of support for precise aeration control systems, resulting in different control capabilities. An ideal blower for precise aeration control should possess the following characteristics:

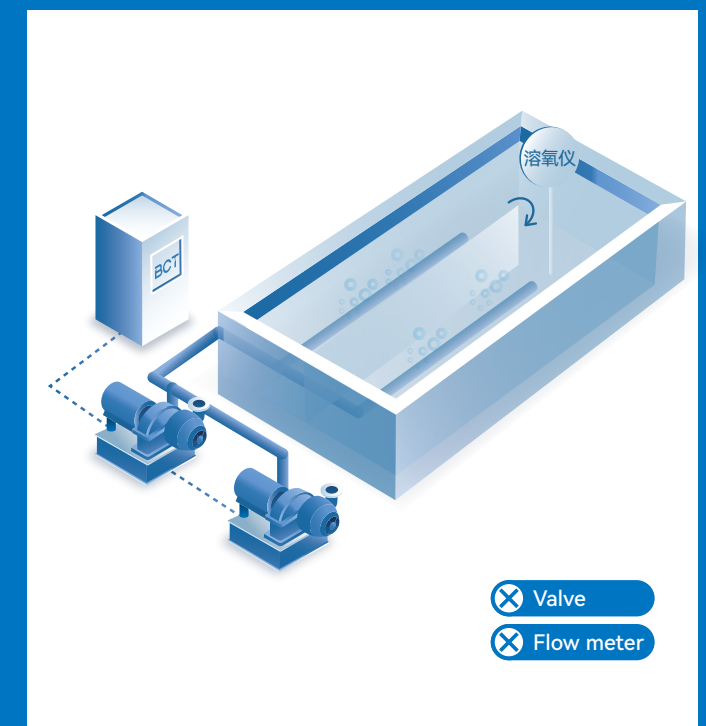
- Robust and reliable performance: Ensures stable operation under various conditions with a low failure rate.
- Wide flow adjustment range: Offers a 45%-100% adjustment range, with consistent and continuous output when multiple blowers operate simultaneously.
- High efficiency throughout the adjustment range: Maintains high energy efficiency across the entire adjustment range, with a linear relationship between specific power and adjustment range.

BCT-Light

For WWTPs with smaller scales, lacking electric valves on the aeration pipelines, or without proper installation positions for flow meters on the aeration pipelines, GPAC® has developed BCT-Light. When paired with a compatible blower, BCT-Light can effectively maintain stable dissolved oxygen levels in the aeration tank.

By eliminating the need for valve control and flow meters, BCT-Light minimizes the potential impact of instrument instability or failure on the reliability of the precise aeration control system.

The Chongqing Tongliang WWTP has been successfully utilizing BCT-Light since 2014. The system has demonstrated stable operation for nearly a decade, achieving a remarkable control accuracy exceeding 99%.



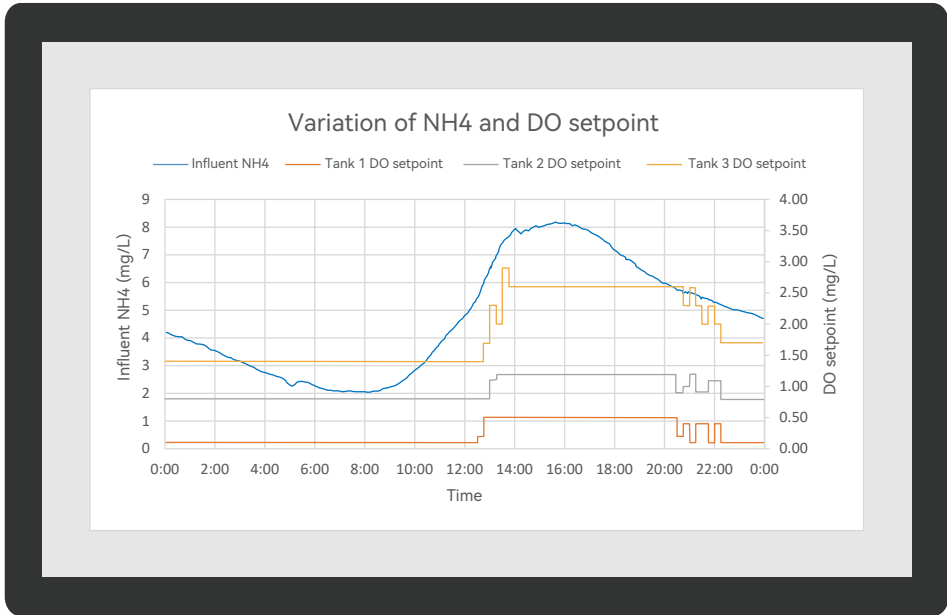
Delivering 90% of the performance at just 10% of the cost!

DYNAMIC DO SETPOINTS

The GPAC® feedforward control system, which is based on the ASM model, employs an iterative approach to assess the impact of various dissolved oxygen target control values on effluent water quality. By analyzing the results, the system identifies the lowest dissolved oxygen value that still ensures compliance with the discharge standard and sets this value as the target control point.

By strategically lowering the dissolved oxygen (DO) setpoint to the minimum level that still guarantees effluent, the demand for blower flow during the aeration process is reduced. This optimization ultimately leads to a decrease in blower energy consumption, promoting energy efficiency without compromising treatment performance.

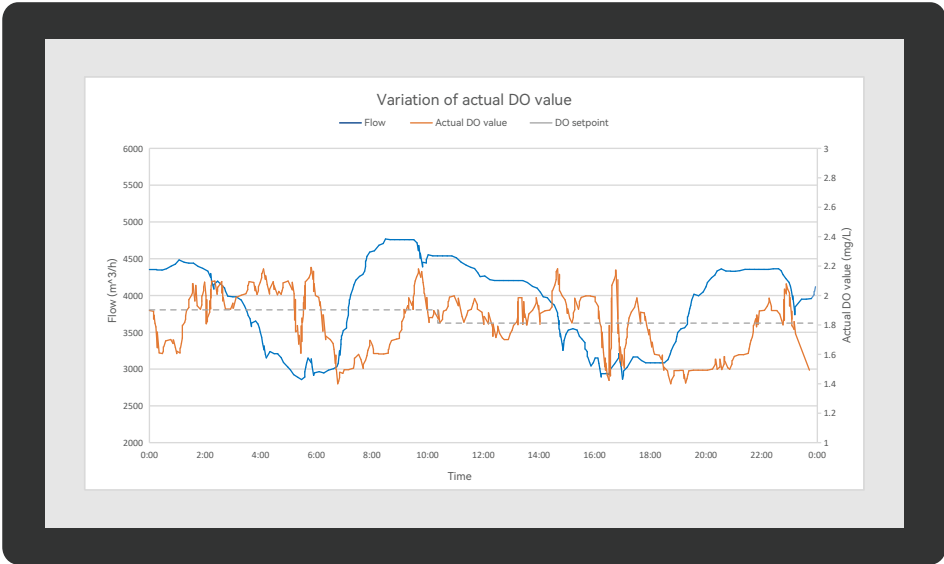
Compared to the prediction lag and frequent adjustment issues associated with traditional PI control, GPAC® incorporates model analysis and big data prediction. The model predicts changes in dissolved oxygen values in advance and adjusts the blower opening accordingly. It then waits for the biochemical reaction in the aerobic tank to stabilize before performing a second adjustment. This approach significantly reduces the oscillation of dissolved oxygen values in the aerobic tank and minimizes equipment wear.



From Denver WWTP, Colorado, USA
GPAC® dynamically adjusts the target control value for dissolved oxygen in the aeration tank based on varying influent loads.

MANUAL DO SETPOINTS

GPAC® also enables users to manually set the target control value for dissolved oxygen based on their practical experience. This feature is especially useful for smaller-scale plants with relatively stable influent load variations.

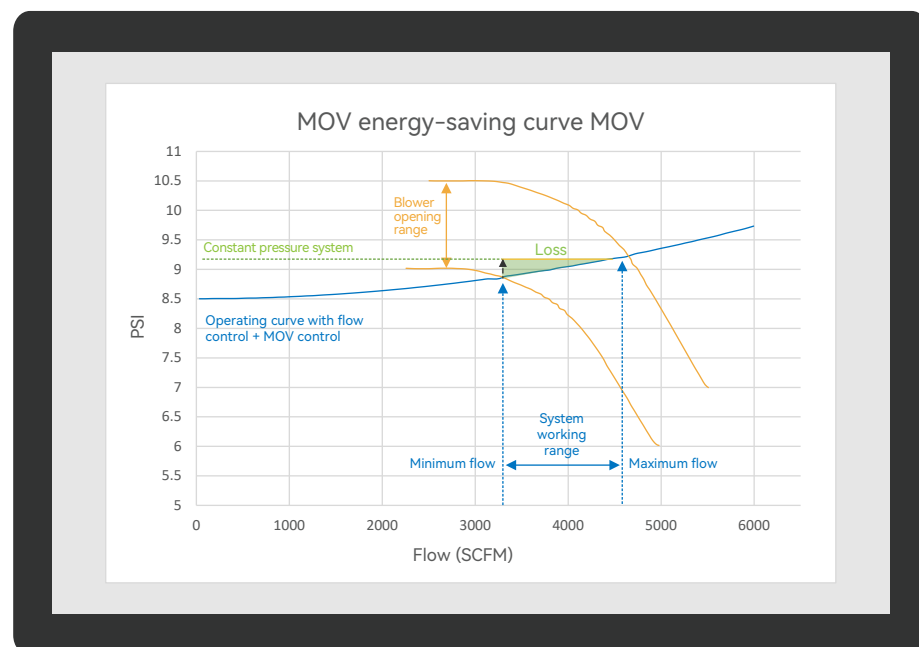


From Chongqing Tongliang WWTP
When the DO setpoint changed from 1.9mg/L to 1.8mg/L, the average flow decreased from 4,089m³/h to 3,846m³/h, saving 5.95% of the air volume.

VALVE CONTROL MODULE (MOV)

Early on, GPAC® recognized that in addition to blower control and dissolved oxygen setting, valve control also has significant energy-saving potential. GPAC® was the first in the industry to research, develop, and implement dynamic maximum opening valve (MOV) control logic as part of its precise aeration control system.

The MOV control module first calculates the flow of each branch pipe and identifies the valve on the branch with the maximum flow as the maximum opening valve (MOV). It then opens that valve to the maximum, thereby reducing the pressure of the entire pipeline system to the minimum and maintaining it at that level. Under this minimum pressure, the module adjusts the opening of the valves on other branches to meet their respective flow requirements. During actual operation, if a branch with a non-MOV valve requires the maximum air volume, that valve becomes the new MOV and opens to the maximum, while the opening of the original MOV is correspondingly reduced.



Compared to non-MOV control, MOV control can save about 4% of aeration energy consumption.

PRECISION DOSING

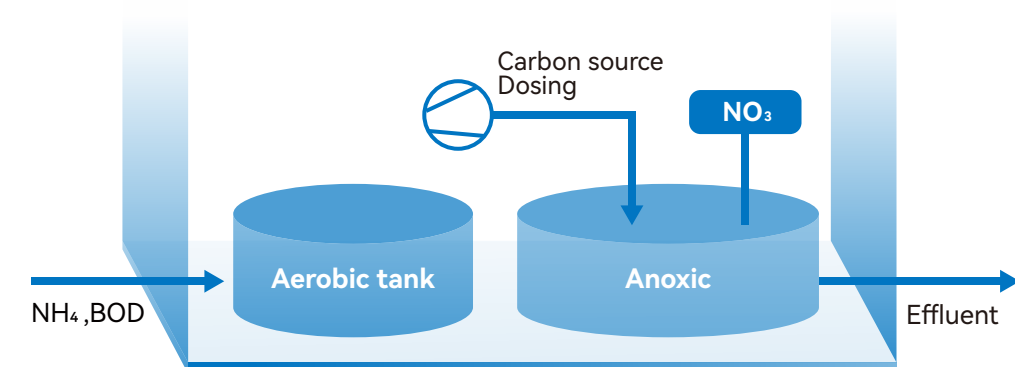
The precise chemical dosing module includes phosphorus removal dosing and carbon source addition.

Phosphorus Removal Dosing Module

GPAC®'s intelligent phosphorus removal dosing system utilizes innovative technology, incorporating well-established mathematical models to analyze biological phosphorus removal efficiency. The system optimizes the calculation of chemical dosage based on the influent total phosphorus load.

The system employs a total phosphorus analyzer installed at the effluent end to monitor water quality in real-time, forming a closed-loop feedback mechanism that continuously corrects model parameters. By interfacing with the automatic dosing device, the system precisely controls the addition of chemicals such as PAC and PAM. This approach reduces chemical consumption and the generation of chemical sludge while ensuring effluent compliance and minimizing overall chemical usage.

Carbon Source Dosing Module



GPAC®'s intelligent carbon source dosing system collects real-time data from the biological tank, including effluent ammonia nitrogen, nitrate nitrogen, internal recirculation, and influent flow, to fit the denitrification rate. The system calculates the carbon source dosing setpoint in real-time and dynamically adjusts the flow of the carbon source dosing pump to achieve on-demand dosing. Additionally, the system employs an effluent total nitrogen meter for feedback adjustment to ensure optimal denitrification effects.

INTELLECTUAL PROPERTY

Continuous innovation through research and development is deeply ingrained in GPAC®'s DNA. Throughout the R&D process, GPAC® has acquired valuable proprietary technologies and core patents covering the essential software and hardware in the field of wastewater treatment automation control. By maintaining its core competitiveness, GPAC® can provide superior products and services to its customers.

No.	Patent No.	Application Date	Approval Date	Patent Name
1	4427778	1982.06.29	1984.01.24	Enzymatic preparation of particulate cellulose for tablet making
2	4803362	2010.01.26	2012.11.13	Optical probe mounting device
3	5401412	1993.07.20	1995.03.28	Method and apparatus for monitoring biological activity in wastewater and controlling the treatment thereof
4	5466604	1994.03.08	1995.11.14	Apparatus for monitoring biological activity in wastewater and controlling the treatment thereof
5	5552319	1994.05.25	1996.09.03	Apparatus and method for monitoring and controlling biological activity in wastewater and controlling the treatment thereof
6	5856119	1996.01.22	1999.01.05	Method for monitoring and controlling biological activity in fluids
7	5698412	1996.11.07	1997.12.16	Method for monitoring and controlling biological activity in fluids
8	6143246	1998.08.18	2000.11.07	Apparatus for measuring ammonia in biochemical processes
9	5976888	1998.12.07	1999.11.02	Method for measuring NO.sub.x in biochemical processes
10	6248595	1999.04.23	2001.06.19	Method of measuring a nitrification rate for liquids
11	6106718	1999.06.30	2000.08.22	Enhanced denitrification process by monitoring and controlling carbonaceous nutrient addition
12	6706171	2001.10.09	2004.03.16	Systems for treating wastewater in a series of filter-containing tanks
13	8308947	2010.01.26	2012.11.13	Methods of providing an aerobic medium in a wastewater treatment bioreactor compartment
14	6706171	2001.10.09	2004.03.16	Systems for treating wastewater in a series of filter-containing tanks
15	63588293	2023.10.06	Pending	An integrated valve and thermal mass flowmeter that measures flow accurately

NOTABLE INSTALLTIONS

Since 2004, GPAC®'s precise aeration control system has been widely implemented in hundreds of projects across more than 20 major countries and regions, including the United States, China, and India. Each project is operating stably, demonstrating clear control and energy-saving benefits.

Time	Project Name	Address	Country	Scale (10,000 tons/day)	Process
2004	Enfield WPCF	Enfield, CT	U.S.A	4	MLE
2005	Zhuyuan WWTP	Shanghai	China	50	AAO
2005	Bailonggang WWTP Phase I	Shanghai	China	120	AAO
2007	Bailonggang WWTP Phase II	Shanghai	China	120	AAO
2008	Phoenix 23rd Ave WWTP	Phoenix, AZ	U.S.A	25	MLE
2010	Yangzhou Liuwei Phase II	Jiangsu	China	10	AAO
2011	Denver MWRD North Secondary	Denver, CO	U.S.A	53	MLE
2012	Putnam WWTP	Putnam, CT	U.S.A	2	MLE
2014	Lebanon WWTP	Lebanon, PA	U.S.A	12	IFAS, AAO
2014	Tongliang WWTP	Chongqing	China	5	AAO
2016	Boitekong WWTP	Rustenburg	South Africa	5	AAO
2017	Wuxi Gaoxin Water New City Phase IV	Jiangsu	China	17	MSBR
2018	TOHO South Bermuda WRF	Kissimmee, FL	U.S.A	8	AAO
2018	Pleasant Hills WWTP	Pleasant Hills, PA	U.S.A	2	MLE
2018	EI Senbellawein WWTP	Cairo	Eqypt	5.5	AAO
2018	Chisinau WWTP	Chisinau	Moldova	34	AAO
2018	Wuxi Huishan Water Treatment Co., Ltd.	Jiangsu	China	2.5	MBR
2019	Wuxi Gaoxin Water New City Plant No. 2	Jiangsu	China	10	BAF
2019	Wuxi Taihu New City WWTP	Jiangsu	China	15	AAO
2019	Wuxi Lucun WWTP	Jiangsu	China	30	AAO
2020	Wuxi Mashan WWTP	Jiangsu	China	3	MBR
2021	Coronation Pillar STP	New Delhi	India	32	AAO
2023	Wuxi Gaoxin Water Meicun Phase IV	Jiangsu	China	5	MSBR
2023	Rochester WWTP	Rochester, NH	U.S.A	2	MLE
2023	Wuxi Chengbei WWTP	Jiangsu	China	22	MBR